CHAPTER 7

GCD, Bézout's and Gauss's theorems

Euclid's algorithm, coprime integers, Diophantine equations

Objectives. Find a gcd with Euclid's algorithm; obtain a Bézout pair; use Bézout's and Gauss's theorems; find an inverse modulo n; solve an equation $ax + by = c$ in integers.

Prerequisites. Chapter 6 (divisibility, Euclidean division, congruences); programming a loop in Python.

Fluency

Fluency · No calculator, five minutes.

1. Carry out the Euclidean division of 1,053 by 325.
2. Give the greatest common divisor of 84 and 126.
3. Do the numbers 15 and 28 have a common divisor other than 1?
4. Find two integers u and v such that $5u + 3v = 1$.
5. Solve $4x \equiv 0 \pmod 6$ for $x \in \{0, \dots, 5\}$.
6. If 7 divides $5k$, what can be conjectured about k?